

MANIPAL
ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES
(A Constituent unit of MAHE, Manipal)

Linux OS and Scripting – 6_Theory

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1. Selected Cybersecurity Scenario

The selected scenario is offline security-log monitoring. A security analyst reviews records for failed login attempts, port scanning, malware alerts and severity levels. Repeated failed logins may indicate password guessing, port scanning may indicate network reconnaissance, and malware alerts may indicate a potentially compromised host. Detecting these events helps prioritize investigation and response.

2. How the Log File Was Obtained

The file security_log.csv was supplied as an attachment for this assignment. Its original publisher, public dataset URL, access date, or simulation procedure was not supplied. Generated using LLM

3. Log File Structure

Field	AWK column	Meaning
Time	$1	Time at which the event was recorded
Source_IP	$2	IP address associated with the event
Event	$3	Event name, such as Login_Failed or Port_Scan
Severity	$4	LOW, MEDIUM, HIGH or CRITICAL
Attempts	$5	Number of attempts associated with the record

4. Cybersecurity Investigation Questions

1. Which Login_Failed records have five or more attempts?
2. Which records have HIGH or CRITICAL severity?
3. How many records belong to each severity category?
4. Which source IP addresses appear more than once?
5. What is the estimated review time if each attempt requires two seconds?

5. Linux Tools Used

- grep – Filters lines by a matching pattern; used to skip the header and select HIGH/CRITICAL rows.
- cut – Extracts selected comma-separated columns, such as severity and source IP.
- sort – Sorts values so identical values are next to each other.
- uniq -c – Counts consecutive repeated values after sorting.
- wc -l – Counts the number of data lines.
- sed – Replaces the event name Login_Failed with Failed_Login in generated output.
- awk – Checks conditions, counts severity categories, identifies suspicious logins and calculates review time.
- Bash – Runs the tools in sequence, creates the output directory and checks for errors.

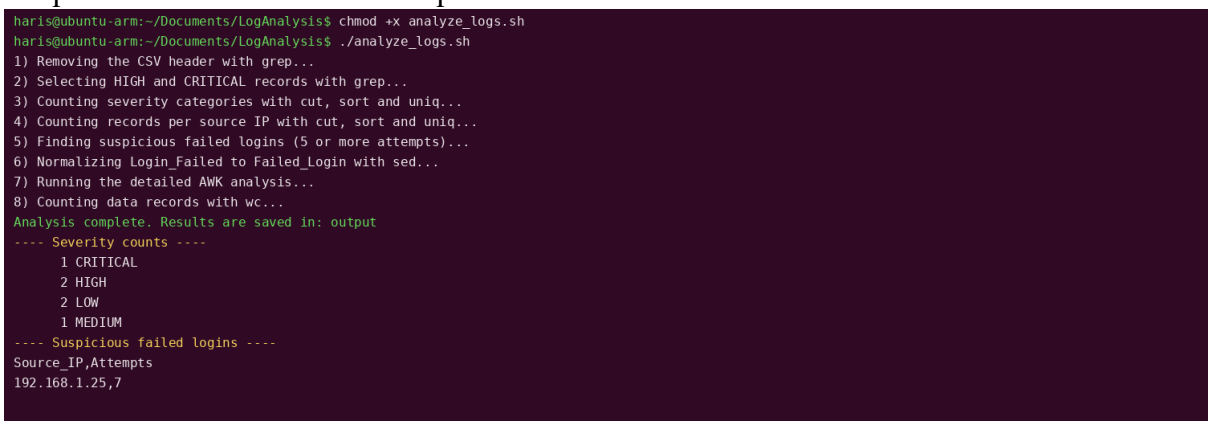
6. Investigation Thresholds

A Login_Failed record is marked suspicious when Attempts is greater than or equal to 5. This threshold follows the assignment's stated example and is used as a simple triage rule, not proof of an attack. HIGH and CRITICAL records are highlighted because the sample log already assigns those severity labels. Review time is calculated as Attempts × 2 seconds, using the assignment's assumption.

7. Results and Suspicious Records

Measure	Result
Total data records	6
Total attempts	25
LOW severity	2
MEDIUM severity	1
HIGH severity	2
CRITICAL severity	1
Suspicious failed-login entries	1
Total estimated review time	50 seconds

The suspicious failed-login rule identifies 192.168.1.25 with 7 attempts. A HIGH-severity Port_Scan record is associated with 10.0.0.8 and 12 attempts. A CRITICAL Malware_Alert is associated with 10.0.0.12. These records should be reviewed first, but the log alone does not prove malicious intent or compromise.



```
haris@ubuntu-arm:~/Documents/LogAnalysis$ cat output/analysis.txt
Entry: 192.168.1.10     Event: Login_Success   Attempts: 1
  Review:  2 seconds
Entry: 192.168.1.25     Event: Login_Failed    Attempts: 3
  Review:  6 seconds
Entry: 10.0.0.8         Event: Port_Scan       Attempts: 12
 Review: 24 seconds
SUSPICIOUS LOGIN: IP=192.168.1.25, attempts=7
Entry: 192.168.1.25     Event: Login_Failed    Attempts: 7
  Review: 14 seconds
Entry: 10.0.0.12        Event: Malware_Alert   Attempts: 1
  Review:  2 seconds
Entry: 192.168.1.30     Event: Login_Success   Attempts: 1   Review:  2 seconds

===== INCIDENT SUMMARY =====
CRITICAL: 1
LOW: 2
MEDIUM: 1
HIGH: 2
Total records: 6
Total attempts: 25
HIGH/CRITICAL records: 3
Suspicious failed-login entries: 1
Total review time: 50 seconds
```

```
haris@ubuntu-arm:~/Documents/LogAnalysis$ cat normalize.sed
# Replace only the exact event field Login_Failed with Failed_Login.
s/,Login_Failed,/,Failed_Login,/g
haris@ubuntu-arm:~/Documents/LogAnalysis$ cat output/normalized.csv
Time,Source_IP,Event,Severity,Attempts
09:10,192.168.1.10,Login_Success,LOW,1
09:12,192.168.1.25,Failed_Login,MEDIUM,3
09:15,10.0.0.8,Port_Scan,HIGH,12
09:18,192.168.1.25,Failed_Login,HIGH,7
09:22,10.0.0.12,Malware_Alert,CRITICAL,1
09:25,192.168.1.30,Login_Success,LOW,1
```

```
haris@ubuntu-arm:~/Documents/LogAnalysis$ cat output/severity_counts.txt
      1 CRITICAL
      2 HIGH
      2 LOW
      1 MEDIUM
haris@ubuntu-arm:~/Documents/LogAnalysis$ cat output/ip_counts.txt
      2 192.168.1.25
      1 192.168.1.30
      1 192.168.1.10
      1 10.0.0.8
      1 10.0.0.12
haris@ubuntu-arm:~/Documents/LogAnalysis$ wc -l output/data_only.csv
6 output/data_only.csv
```

8. False Positives and Limitations

A legitimate user may enter an incorrect password several times. A port scan may be part of an authorized inventory or security assessment. A malware alert may be a false detection. The dataset is small and has limited context: it does not include usernames, destination ports, hostnames, event IDs, packet details or a long time range. Attempts may be aggregated counts rather than individual events. The scripts assume simple CSV fields without commas inside quoted values.

9. Conclusion

The assignment demonstrates how Bash, grep, cut, sort, uniq, wc, sed and awk can be combined for a repeatable first-pass log review. The main investigation leads are the HIGH port-scan record, the HIGH failed-login record with seven attempts, and the CRITICAL malware alert. Further authorized investigation would be needed to confirm any incident.